

P-11

SESSION (2024 – 25)

PRE BOARD – I

CLASS: X

SUB: MATHEMATICS (BASICS)

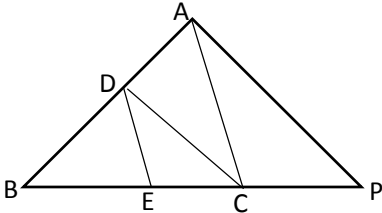
TIME: 3 Hours

M.M: 80

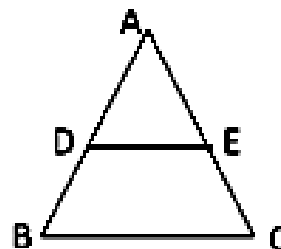
GENERAL INSTRUCTIONS:

1. This question paper contains 38 questions.
2. This question paper is divided into 5 sections A, B, C, D and E.
3. In Section A, questions no. 1 to 18 are multiple choice questions (MCQs) and question no 19 and 20 are Assertion- Reason based questions of 1 mark each.
4. In Section B, questions no. 21 to 25 are very short answer (VSA) type questions, carrying 2 marks each.
5. In Section C, questions no. 26 to 31 are short answer (SA) type questions, carrying 3 marks each.
6. In Section D, questions no. 32 to 35 are long answer (LA) type questions, carrying 5 marks each.
7. In Section E, questions no. 36 to 38 are case study based questions carrying 4 marks each with sub parts of the values of 1, 1 and 2 marks each respectively.
8. All questions are compulsory. However, an internal choice in 2 questions of section B, 2 questions of section C and 2 questions of section D has been provided. And internal choice has been provided in all the 2 marks questions of Section E.

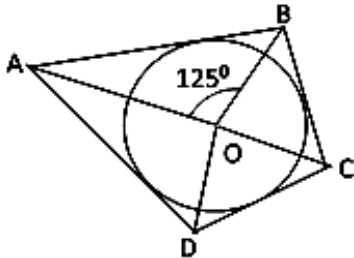
SECTION A (1 X 20 = 20)		
Section A consists of 20 questions of 1 mark each.		
1.	If the quadratic equation is a perfect square then discriminant is: a) Equal to zero b) Less than zero.	1

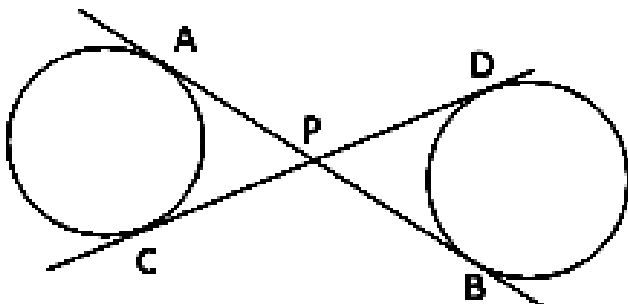
	c) Greater than zero. d) Greater than or equal to zero	
2.	If n^{th} term of an A.P is $(2 - 3n)$ then common difference is : a) (-3) b) 3 c) 0 d) (-1)	1
3.	In the given figure, $DE \parallel AC$, $CD \parallel AP$, which of the following statement is not correct: a) $\frac{BE}{EC} = \frac{BC}{CP}$ b) $(BE) \times (CP) = (EC) \times (BC)$ c) $\frac{EC}{EB} = \frac{CP}{BC}$ d) $(BE) \times (BC) = (EC) \times (PC)$ 	1
4.	The distance of $(3,4)$ from the mid-point of $(6, 8)$ and $(2, -4)$ is: a) 25 b) 12 c) $5\sqrt{2}$ d) $\sqrt{5}$	1
5.	If the distance between two parallel tangents is 18 cm then the radius of the circle is: a) 7 cm b) 9 cm c) 8 cm d) 1.8 cm	1
6.	Find the value of θ , if $\sin \theta = \frac{1}{2}$ a) 0° b) 30°	1

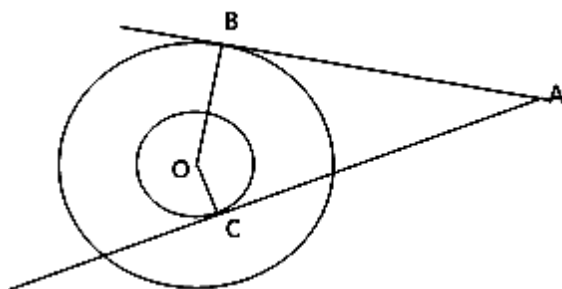
	c) 60° d) 45°	
7.	If mean = (3 median – mode) $\times k$, then the value of k is: a) 1 b) 2 c) $\frac{1}{2}$ d) $\frac{3}{2}$	1
8.	The sum of the probabilities of all elementary events of an experiment is p, then a) $0 < p < 1$ b) $0 \leq p < 1$ c) $p = 1$ d) $p = 0$	1
9.	If P(x) is a polynomial of at least degree one and P(k) = 0, then k is known as a) Value of P(x) b) Zero of P(x) c) Constant term of p(x) d) None of these	1
10.	A cylinder, cone and hemisphere are of same base and of same height, the ratio of their volumes is: a) 1: 2: 3 b) 2: 1: 3 c) 3: 1: 2 d) 3: 2: 1	1
11.	In the given figure, $\frac{AD}{BD} = \frac{AE}{EC}$ and $\angle ADE = 70^\circ$, $\angle BAC = 50^\circ$, then $\angle BCA =$ a) 70° b) 50° c) 80° d) 60°	1
12.	The ratio between the L.C.M and H.C.F of 5, 15, 20 is:	1

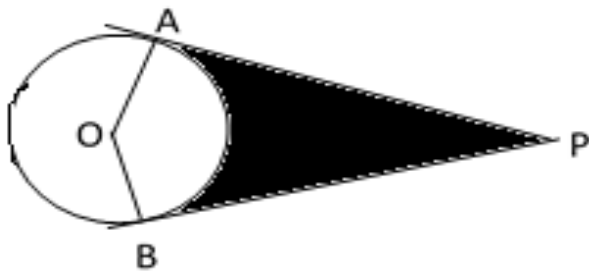


	a) 9 : 1 b) 4 : 3 c) 11 : 1 d) 12 : 1													
13.	Match the column: <table><tr><td>1</td><td>$2x + 5y = 10$ $3x + 4y = 7$</td><td>A</td><td>Unique Solution</td></tr><tr><td>2</td><td>$2x + 5y = 10$ $6x + 15y = 20$</td><td>B</td><td>Infinitely many solution</td></tr><tr><td>3</td><td>$5x + 2y = 10$ $10x + 4y = 20$</td><td>C</td><td>No common solution</td></tr></table> a) 1 – A , 2 – B , 3 – C b) 1 – B , 2 – C , 3 – A c) 1 – C , 2 – B , 3 – A d) 1 – A , 2 – C , 3 – B	1	$2x + 5y = 10$ $3x + 4y = 7$	A	Unique Solution	2	$2x + 5y = 10$ $6x + 15y = 20$	B	Infinitely many solution	3	$5x + 2y = 10$ $10x + 4y = 20$	C	No common solution	1
1	$2x + 5y = 10$ $3x + 4y = 7$	A	Unique Solution											
2	$2x + 5y = 10$ $6x + 15y = 20$	B	Infinitely many solution											
3	$5x + 2y = 10$ $10x + 4y = 20$	C	No common solution											
14.	Which of the following equation has the sum of its roots as 3? a) $2x^2 - 3x + 6 = 0$ b) $-x^2 + 3x - 3 = 0$ c) $\sqrt{2}x^2 - \frac{3}{\sqrt{2}} + 1 = 0$ d) $3x^2 - 3x + 3 = 0$	1												
15.	In ΔABC , $\angle B = 90^\circ$, $AB = 5\text{ cm}$, $\angle ACB = 30^\circ$ then $BC =$ a) $5\sqrt{3}\text{ cm}$ b) $2\sqrt{3}\text{ cm}$ c) 10 cm d) $10\sqrt{3}\text{ cm}$	1												
16.	If p is prime, then H.C.F and L.C.M of p and $(p + 1)$ would be a) H.C.F = p and L.C.M = $(p + 1)$ b) H.C.F = $p(p + 1)$ and L.C.M = 1 c) H.C.F = 1 and L.C.M = $p(p + 1)$ d) None of these.	1												

17.	In the given figure, if $\angle AOB = 125^\circ$ then $\angle COD =$ a) 62.5° b) 45° c) 35° d) 55° 	1
18.	The height of mountains is found out using the idea of indirect measurements which is based on the a) Principle of congruent figure. b) Principle of similarity of figure. c) Principle of Equality of figures. d) None of these	1
	Direction: In the question number 19 and 20, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option: (A) Both assertion (A) and Reason (R) are true and reason (R) is the correct explanation of assertion (A). (B) Both assertion (A) and Reason (R) are true and reason (R) is not the correct explanation of assertion (A). (C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.	
19.	Assertion (A): $3 \times 5 \times 7 + 7$ is a composite number. Reason (R): A composite number has factors one itself and any other natural number.	1
20.	Assertion: The sum of 15 terms of the A.P 1, 3, 5,, is 225. Reason: The sum of first n odd natural numbers is n^2.	1
SECTION B (2 X 5 = 10)		
Section B consists of 5 questions of 2 mark each.		

21.	If the common difference of an A.P is 5 then find $a_{18} - a_{13}$.						2
22.	For the following distribution:						2
	Marks	Below 10	Below 20	Below 30	Below 40	Below 50	Below 60
	No. of students	3	12	27	57	75	80
	Find the median class of the data.						
23.	Find the ratio in which the line segment joining (- 2, - 3) and (5, 6) is divided by $x - axis$. OR Find a relation between x and y such that the point (x, y) is equidistant from (2, 5) and (- 3, 7).						2
24.	Evaluate: $\frac{4}{3} Cot^2 30^\circ + \frac{1}{3} Sec^2 60^\circ - \frac{3}{4} Cosec^2 30^\circ$. OR In $\Delta ABC, \angle A = 90^\circ$. If $\sin B = \frac{21}{29}$, find $Cos C$						2
25.	In the given figure, common tangents AB and CD to two circles intersect at P. Prove that: $AB = CD$. 						2
SECTION C (3 X 6 = 18)							
Section C consists of 6 questions of 3 mark each.							
26.	The sum of fourth and eighth terms of an A.P is 24 and the sum of the sixth and tenth term is 44. Find the first term and the common difference of the A.P. OR						3

	Find the sum of first 25 terms of an A.P whose nth term is given by $a_n = 2 - 3n$.													
27.	Find the perimeter of the triangle having vertices (0, 3), (4, 0) and (0, 0).	3												
28.	If the tangents are drawn to a circle from an external point then prove that they subtend equal angle at the centre. OR Two concentric circles are of radii 10cm and 8cm. AB and AC are tangents to two circles from A. If AB = 24 cm then find AC. 	3												
29.	Prove that: $\frac{1+\sec\theta}{\sec\theta} = \frac{\sin^2\theta}{1-\cos\theta}$	3												
30.	a) State the “Fundamental Theorem of Arithmetic.” b) Find the H.C.F of 96 and 404 by the prime factorization method. Hence find their L.C.M.	1 2												
31.	Find the mode of the following distribution:	3												
I	<table><tr><td>Class</td><td>100 - 120</td><td>120 - 140</td><td>140 - 160</td><td>160 - 180</td><td>180 - 200</td></tr><tr><td>Frequency</td><td>12</td><td>14</td><td>8</td><td>6</td><td>10</td></tr></table>	Class	100 - 120	120 - 140	140 - 160	160 - 180	180 - 200	Frequency	12	14	8	6	10	
Class	100 - 120	120 - 140	140 - 160	160 - 180	180 - 200									
Frequency	12	14	8	6	10									
SECTION D (5 X 4 = 20)														
Section D consists of 4 questions of 5 mark each.														
32.	The angle of elevation and depression of the top and bottom of a light house from the top of a 60 m high building are 30° and 60° respectively. Find: a) The difference between the height of the light house and the building. b) The distance between the light house and the building. OR	5												

	The angle of elevation of an aeroplane from a point A on the ground is 60° . After a flight of 15 seconds, the angle of elevation changes to 30° . If the aeroplane is flying at a constant height of $1500\sqrt{3} \text{ m}$, find the speed of the plane in km/h.	
33.	Find the values of a and b for which the following pair of linear equations has infinitely many solutions: $2x + 3y = 7$ $(a + b)x + (2a - b)y = 21$ OR Draw the graphs of the equations $x - y + 1 = 0$ and $3x + 2y - 12 = 0$. Determine the coordinate of the vertices of the triangle formed by these lines and the x-axis and shade the triangular region.	5
34.	a) "If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct point the other two sides are divided in the same ratio" Prove it. b) E is a point on the side AD produced of a parallelogram BE intersects CD at F show that $\triangle ABE \sim \triangle CFB$.	3 2
35.	An elastic belt is placed round the rim of a pulley of radius 5 cm. One point on the belt is pulled directly away from the centre O of the pulley until it is at P, 10 cm away from O. Find the length of the belt that is in contact with the rim of the pulley. Also find the shaded area. (Use $\pi = 3.14, \sqrt{3} = 1.73$). 	5
SECTION E (4 X 3 = 12)		

36. Adventure camps are the perfect place for the children to practice decision making for themselves without parents and teachers guiding their every move. Some students of a school reached for adventure at Sakleshpur. At the camp, the waiters served some students with a welcome drink in a cylindrical glass and some students in a hemispherical cup whose dimensions are shown below. After that they went for a jungle trek. The jungle trek was enjoyable but tiring. As dusk fell, it was time to take shelter. Each group of four students was given a canvas of area 551 m^2 . Each group had to make a conical tent to accommodate all the four students. Assuming that all the stitching and wasting incurred while cutting, would amount to 1 m^2 , the students put the tents. The radius of the tent is 7 m .

- Find the volume of the cylindrical cup.
- Find the volume of the hemispherical cup.
- Find the height of the conical tent prepared to accommodate four students.

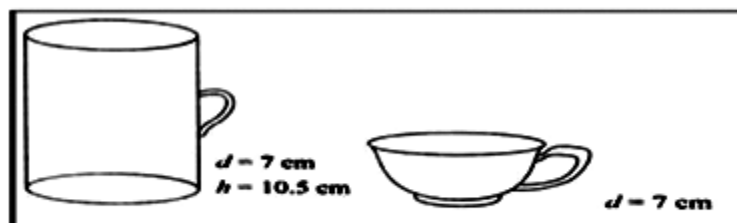
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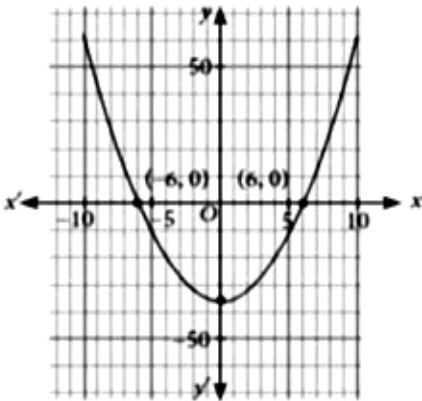
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2

OR

How much space on the ground is occupied by each student in the conical tent?



37.	While visiting his village, Sohan saw a honeycomb and asked his father what is that. He replied that it's a honeycomb made by honey bees to store honey. Also, he told him that the shape of the honeycomb formed is parabolic. The mathematical representation of the honeycomb structure is shown in the graph. a) What are the zeroes of the polynomial? b) Find the expression of the polynomial represented by the graph. OR Verify the relationship between the zeroes and the coefficients. c) What is the degree of the polynomial?  	1 2 1
38	Two friend Dhoni and Sachin are travelling in a train. They feeling bored, so they started played a game with a pair of dice that one of them had, each of them started rolling the pair of dice one by one, starting one condition before rolling. If the person gets the numbers according to the condition stated by him, he wins and get a score. a) Dhoni says, "a doublet". What is the probability of his winning? b) Sachin says, ∴ sum less than 9. What is the probability of his winning?	1 1

	c) Dhoni says, “6 will come up either time”. What is the probability of his winning? OR Sachin says, ∴ sum is an even number. What is the probability of his losing?  Two dice are shown. The top die shows 1, 2, and 3. The bottom die shows 4, 5, and 6.	2
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